Ashwitha.R

2303A51340

SQL

1

```
SELECT d.dept_name, COUNT(e.emp_id) AS employee_count, AVG(e.salary) AS avg_salary, MIN(e.salary) AS min_salary, MAX(e.salary) AS max_salary FROM Departments d LEFT JOIN Employees e ON d.dept_id=e.dept_id GROUP BY d.dept_id,d.dept_name;
```

2

```
SELECT emp_id,emp_name,dept_id,salary FROM (SELECT *,DENSE_RANK() OVER(PARTITION BY dept_id ORDER BY salary DESC) AS r FROM Employees) s WHERE r=2;
```

3

```
SELECT emp_id,emp_name,dept_id,salary FROM Employees e WHERE salary>(SELECT AVG(salary) FROM Employees WHERE dept_id=e.dept_id);
```

4

```
SELECT m.emp_name,m.salary AS manager_salary,COUNT(e.emp_id) AS team_count,SUM(e.salary) AS team_salary FROM Employees m JOIN Employees e ON m.emp_id=e.manager_id GROUP BY m.emp_id,m.emp_name,m.salary HAVING COUNT(e.emp_id)>=2;
```

5

```
SELECT project_id,emp_id,total_hours FROM (SELECT project_id,emp_id,SUM(hours_worked) AS total_hours,DENSE_RANK() OVER(PARTITION BY project_id ORDER BY SUM(hours_worked) DESC) AS r FROM EmployeeProjects GROUP BY project_id,emp_id) s WHERE r<=2;
```

6

```
SELECT dept_id,dept_name FROM Departments d WHERE EXISTS(SELECT 1 FROM Employees e WHERE e.dept_id=d.dept_id) AND NOT EXISTS(SELECT 1 FROM Projects p WHERE p.dept_id=d.dept_id);
```

7

```
SELECT order_date,SUM(amount) AS daily_revenue,SUM(SUM(amount)) OVER(ORDER BY order_date) AS total_revenue FROM Orders WHERE status='completed' GROUP BY order_date ORDER BY order_date;
```

8

```
SELECT customer_id,SUM(amount) AS total_value,CASE WHEN SUM(amount)>=100000 THEN
'PLATINUM' WHEN SUM(amount)>=50000 THEN 'GOLD' WHEN SUM(amount)>=20000 THEN 'SILVER'
ELSE 'BRONZE' END AS category FROM Orders WHERE status='completed' GROUP BY customer_id;
```

9

```
WITH M AS (SELECT DISTINCT customer_id,DATE_TRUNC('month',order_date) AS mon FROM Orders),N
AS (SELECT customer_id,mon,LAG(mon,1) OVER(PARTITION BY customer_id ORDER BY mon) AS
p1,LAG(mon,2) OVER(PARTITION BY customer_id ORDER BY mon) AS p2 FROM M) SELECT DISTINCT
customer_id FROM N WHERE mon=p1+INTERVAL '1 month' AND mon=p2+INTERVAL '2 months';
```

10

```
SELECT dept_id,emp_id,emp_name,salary,CASE WHEN salary=highest THEN 'HIGHEST' WHEN
salary=lowest THEN 'LOWEST' END AS position,highest-lowest AS salary_difference FROM (SELECT
*,MAX(salary) OVER(PARTITION BY dept_id) AS highest,MIN(salary) OVER(PARTITION BY dept_id) AS
lowest FROM Employees) s WHERE salary=highest OR salary=lowest;
```